

a Transform one declared operator

$$\Delta J_{\text{phys}} = \varepsilon I$$

$$\Downarrow D_r^{-1}(\cdot)D_x$$

$$\widehat{\Delta J} = \varepsilon D_r^{-1}D_x$$

Equivalent coordinate representation

b Redeclare after scaling

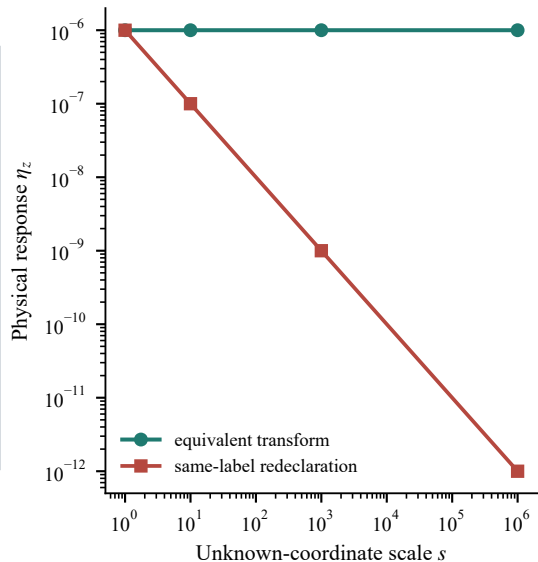
$$\widehat{\Delta J} = \varepsilon I$$

$$\Downarrow D_r(\cdot)D_x^{-1}$$

$$\Delta J_{\text{phys}} = \varepsilon D_r D_x^{-1}$$

A different physical perturbation

c Response to the represented operator



Exact mapping and registered reproduction; no preferred or optimal scaling is inferred